L4 Validation Test Record

Topic: Quantum Harmonic Oscillator

AITP Protocol v3 — L4 Regression Test

2026-04-20

Abstract

This TeX document is a preserved test record of L4 validation output for the quantum harmonic oscillator energy spectrum candidate. It captures the full L4 review structure: six-outcome vocabulary, physics check results, devil's advocate record, and SymPy verification evidence.

Contents

1 Candidate Under Review

Candidate Summary

Candidate ID: `cand-qho-energy-spectrum`
Claim: The energy eigenvalues of the 1D quantum harmonic oscillator with Hamiltonian $H = p^2/(2m) + \frac{1}{2}m\omega^2x^2$ are $E_n = (n + 1/2)\hbar\omega$, for $n = 0, 1, 2, \ldots$
Regime of Validity: Non-relativistic QM, 1D, harmonic potential only.
Type: `result` **Lane:** `formal_theory`

2 L4 Validation Outcome

2.1 Primary Review (Cycle 1)

Review: cand-qho-energy-spectrum

Outcome: PASS `pass`
Review Date: 2026-04-20T12:00:00Z
Notes: All four mandatory physics checks passed. The derivation correctly recovers the well-known quantum harmonic oscillator energy spectrum. The ladder operator method is a standard technique validated across multiple textbooks.

2.2 Physics Check Results

Check	Result	Detail
Dimensional consistency	PASS	$[H] = [\hbar\omega] =$ energy, both sides ML^2T^{-2}
Symmetry compatibility	PASS	H commutes with parity operator $\hat{P}$
Limiting case check	PASS	Classical HO recovered as $n \to \infty$: $T = 2\pi/\omega$
Correspondence check	PASS	Matches known result $E_n = (n + 1/2)\hbar\omega$ (Griffiths §2.3)

2.3 Devil's Advocate

The derivation assumes the potential is exactly harmonic. Anharmonic corrections (x^3, x^4 terms) would shift the spectrum. Regime boundary: coupling must be weak enough that perturbation theory converges.

2.4 SymPy Verification Evidence

Tool: `aitp_verify_dimensions`


```
{
  "pass": true,
  "lhs_dimension": [1, 2, -2, 0, 0],
```

```
  "rhs_dimension": [1, 2, -2, 0, 0]
}
```


3 Versioned Review (Cycle 2)

Review: cand-qho-energy-spectrum_v2

Outcome: PASS pass
Review Date: 2026-04-21T09:00:00Z
Notes: Second review cycle after addressing v1 feedback on anharmonic corrections. Confirmed that the harmonic spectrum is correct in the weak-coupling limit. Anharmonic corrections estimated at $O(\lambda)$ with $\lambda \ll 1$.

3.1 Updated Check Results (v2)

Check	Result	Detail
Dimensional consistency	PASS	Reconfirmed after anharmonic correction analysis
Symmetry compatibility	PASS	Parity confirmed even with x^4 perturbation
Limiting case check	PASS	Classical HO period $T = 2\pi/\omega$ recovered
Correspondence check	PASS	Matches known result after v1 corrections

3.2 Devil's Advocate (v2)

The anharmonic x^4 correction shifts E_0 by $\sim 0.01\,\hbar\omega$ at weak coupling. Stronger coupling would break the harmonic approximation entirely.

4 Multi-Outcome Summary

This section records the complete six-outcome L4 vocabulary as exercised by the test suite (`test_l4_l2_memory.py`):

Candidate ID	Outcome	Summary
`cand-qho-energy-spectrum`	PASS pass	All checks passed; validated for promotion
`cand-spin-chain-gs`	INCONCLUSIVE partial_pass	Symmetry check inconclusive; partial trust
`cand-incorrect-gap`	FAIL fail	Correspondence check failed (3.2 vs 1.1 eV)
`cand-parity-anomaly`	FAIL contradiction	Parity conservation contradicts Redlich (1984)
`cand-nonperturbative-gap`	INCONCLUSIVE stuck	Perturbation theory inadequate; lattice method unavailable

Candidate ID	Outcome	Summary
cand-hpc-timeout	INCONCLUSIVE timeout	Slurm job 47281 exceeded 48h walltime

5 Trust Classification

After L4 validation, promoted candidates receive 2D trust classification:

Candidate	Trust Basis	Trust Scope
cand-qho-energy-spectrum	validated	bounded_reusable
cand-spin-chain-gs	partial	experimental
cand-incorrect-gap	none	none
cand-parity-anomaly	falsified	none

6 Test Fixture Provenance

- **Fixture directory:** `tests/fixtures/l4/`
- **Review files:** `reviews/cand-*.md` (6 outcomes + 1 versioned)
- **Validation contract:** `validation_contract.md`
- **Physics check baseline:** `physics_check_baseline.json`
- **Candidate baseline:** `candidate_baseline.md`
- **TeX records:** `tex/` (this file and siblings)
- **Test file:** `tests/test_l4_l2_memory.py`